OCTO-EARTH

# Calculation Review Report

Evidence-based review of the orbital, launcher, power-beaming, mirror, climate, and numerical calculations

**EXECUTIVE FINDING**
Most elementary arithmetic checks under the stated assumptions. This report is not evidence of engineering feasibility, safe Earth relocation, climate safety, or successful PPO learning. Key issues include an apparently missing circular-aperture factor in C121, an invalid “total impulse ÷ initial mass” check in C39, omitted photon momentum in C81, and unverified mission-result claims. The arm-speed calculation is conditional on an extreme and ambiguous cross-section assumption.

## At a glance

| Metric | Value in source report | Review status |
|---|---|---|
| **Target / deadline** | 2.7 AU / 10,000 simulated years | Hypothetical mission target, not a real-world recommendation. |
| **Screened slow-spiral Δv** | 11,658 m/s | Ideal quasi-circular screen, not exact finite-mission Δv. |
| **Minimum launch energy** | $1.66\times10^{35}$ J | Conditional on idealized rocket, mass, and efficiency assumptions. |
| **Arm speed screen** | 19,357 km/s static equation limit | Only if geometry and 86.67 GPa allowable are valid. |
| **C121 beam capture at target** | PDF: 2.1% (100 km receiver) | Circular-area estimate: ≈1.31%; see C121 note. |
| **Policy outcomes** | Different values in the two supplied reports | Unreconciled; raw comparison artifacts were not supplied. |

### Mission and claim boundary

The scenario moves the Earth–Moon system outward by ejecting finite asteroid mass and relies on assumed solar collectors, microwave transmission, an Earth launcher, and space mirrors. These are hypothetical enablers. A simulation result is not a design, feasibility proof, safety assessment, or climate/habitability prediction.

The two supplied reports disagree on PPO outcomes. Performance values are retained as attributed claims, not independently verified results.

# 1. What the system is and the three enablers

OCTO-EARTH moves Earth (with its Moon) from its orbit at 1 AU to a circular orbit at **2.7 AU** within **10,000 simulated years** by throwing mass away from Earth (momentum exchange: Earth recoils the other way). The thrown mass comes from an asteroid docked to Earth, the energy comes from sunlight collected near the Sun and beamed to Earth as microwaves, and Earth's climate is kept livable far from the Sun by mirrors that reflect extra sunlight onto it. A reinforcement-learning agent (PPO) chooses every 100 years how hard, how fast and in which direction to throw, and how much mirror light to send to Earth.

**Exactly three enablers.** An enabler is a technology the model ASSUMES can be built without showing it. The model allows only three; each one is a complete system made of the instruments listed under it. Everything else in the model is ordinary physics (gravity, momentum, energy, radiation) or ordinary control software.

| | Enabler | What it does in the model | Instruments |
|---|---|---|---|
| **E1** | Earth launcher system: docked asteroid + launch ring with 8 arms + arm radiators | Throws packets of up to 1e18 kg of asteroid material at up to 5,000 km/s, 8 times per year (all 8 arms throw one packet together), with 50 % of the drawn energy ending up as packet kinetic energy; the arms reject their waste heat away from Earth; the launches do not damage Earth. | • Asteroid reservoir (docked to Earth)<br>• Launch ring + 8 launch arms (mass driver)<br>• Arm power conditioning + energy reserve<br>• Directional arm radiators |
| **E2** | Solar power belt with microwave power beaming (collectors + transmitters + arm rectennas) | 5 million thin-film collectors at 0.5 AU intercept 30 % of the Sun's output, turn it into electricity and beam it as 5.8 GHz microwaves (30 km phased arrays) to 1,000 km rectennas on the 8 arms; they hold their slots by sail tilt, keep a clear sunlight lane for Earth and Mars, keep beam spill off Earth and are built up / replaced by factories using Mercury material. | • Thin-film solar collector (x 5 million)<br>• Microwave phased-array transmitter (one per collector)<br>• Rectenna on each of the 8 arm tips<br>• Sail-tilt station keeping (on each collector)<br>• Clear sunlight lane<br>• Collector factories (Mercury material) |
| **E3** | Space mirror ring around Earth (soletta) | 1.6e15 m^2 of aluminium-film mirrors (8e12 kg) in a terminator orbit 20 Earth radii out reflect extra sunlight onto Earth: up to 1.10 x today's sunlight at 2.7 AU. The agent sets how much (mirrors not needed are turned so their light passes Earth). | • Mirror film panels (aluminium, 5 g/m^2)<br>• Terminator orbit at 20 Earth radii + pointing (45 degree tilt) |

## Chain of cause and effect

Sunlight → **E2** collectors (electricity) → **E2** transmitters (microwaves) → **E2** rectennas on the arms (electricity) → **E1** launch arms throw asteroid packets (**E1** reservoir) → Earth recoils, its orbit grows → less sunlight at Earth → **E3** mirrors add sunlight → climate stays near 288 K. Waste heat of E1 and E2 is sent away from Earth by the E1 radiators and the E2 spill control; the small share that still reaches Earth is part of the climate calculation.

## Key numbers at a glance

| Quantity | Value |
|---|---|
| Required speed change of Earth (slow spiral) | 11,658 m/s |

| | |
|---|---|
| Orbital energy Earth+Moon must gain | $1.70\times10^{33}$ J |
| Minimum launch energy (whole asteroid, slowest launch) | $1.66\times10^{35}$ J |
| Asteroid (reaction mass) | $3.00\times10^{22}$ kg = 0.408 Moon masses |
| Launch speed used by the hand-designed strategy | 2,480 km/s = 0.83 % of light speed |
| Sunlight intercepted by the collector belt | $1.15\times10^{26}$ W (30 % of the Sun) |
| Electricity at the arms (Earth at 1 AU / at the target) | $1.11\times10^{25}$ W (weakest point) |
| Space-mirror capacity at the target | 1.10 × today's sunlight on Earth |
| Analytic verdict | **FEASIBLE within this model (physically hypothetical scale)** |
| Tested mission (hand-designed, high fidelity) | success: a = 2.7000 AU, e = 0.0081 after 3,600 y; surface 288.3 K |

Numbers are labelled **C1, C2, …**. Each shows the formula, the values put in, the result and the function in `octoearth_rl.py` that does the same calculation in the simulation. Values marked *indicative* are NOT used by the simulation (they only help to judge an instrument).

## 2. Physical constants and mission inputs

| Constant | Value | Unit |
|---|---|---|
| Gravitational constant G | $6.67430\times10^{-11}$ | m^3 kg^-1 s^-2 |
| Sun mass | $1.98841\times10^{30}$ | kg |
| Earth mass | $5.97217\times10^{24}$ | kg |
| Moon mass | $7.34579\times10^{22}$ | kg |
| Mars mass | $6.41691\times10^{23}$ | kg |
| Mars radius | $3.38950\times10^{6}$ | m |
| Astronomical unit (AU) | $1.49598\times10^{11}$ | m |
| Year | $3.15576\times10^{7}$ | s |
| Earth radius | $6.37100\times10^{6}$ | m |
| Moon distance | $3.84400\times10^{8}$ | m |
| Solar constant at 1 AU | 1,361.00 | W/m^2 |
| Speed of light | $2.99792\times10^{8}$ | m/s |

| Mission input | Value |
|---|---|
| Target orbit radius $a_t$ | 2.7 AU (tolerance ±0.05 AU, e ≤ 0.05, i ≤ 5.0°) |
| Mission time limit T | 10,000 years = $3.16\times10^{11}$ s |
| Decision interval | 100 years (100 decisions) |
| Climate target | 288 K ± 5 K |

The full list of all parameters with their values, units and meaning is in the appendix.

# 3. Mission requirement: how much push, energy and time

**C1** **Mass that has to be moved**

Formula $m_{sys} = M_{Earth} + m_{asteroid} + M_{Moon}$

Values $5.9722\times10^{24} + 3.00\times10^{22} + 7.346\times10^{22}$ kg

Result **$6.0756\times10^{24}$ kg**

code: feasibility()

**C2** **Gravitational parameter of the Sun-Earth system**

Formula $\mu = G\cdot M_{Sun} + G\cdot m_{sys}$

Values $1.32712\times10^{20} + 4.055\times10^{14}$ m³/s²

Result **$1.32713\times10^{20}$ m³/s²**

code: feasibility()

**C3** **Circular orbital speed today (1 AU)**

Formula $v_0 = \sqrt{(\mu / a_0)}$

Values $\sqrt{(1.327\times10^{20} / 1.4960\times10^{11})}$

Result **29,784.7 m/s**

code: feasibility()

**C4** **Circular orbital speed at the target (2.7 AU)**

Formula $v_t = \sqrt{(\mu / a_t)}$

Values $\sqrt{(1.327\times10^{20} / 4.0391\times10^{11})}$

Result **18,126.4 m/s**

code: feasibility()

**C5** **Required speed change (slow outward spiral)**

Formula $\Delta v = v_0 - v_t$

Values 29,784.7 − 18,126.4

Result **11,658 m/s**

Meaning A slow spiral (small pushes along the motion over thousands of years) needs exactly the difference of the two circular speeds.

code: feasibility() dv_spiral

**C6** **Reference: Hohmann transfer (two short burns)**

Formula $a_{tr} = (a_0+a_t)/2$; $\Delta v = |\sqrt{(\mu(2/a_0-1/a_{tr}))} - v_0| + |v_t - \sqrt{(\mu(2/a_t-1/a_{tr}))}|$

Values $a_{tr}$ = 1.850 AU

Result **10,997 m/s**

Meaning Cheaper in speed change, but would need two enormous short pushes; the launcher can only push gradually, so the spiral value is used.

code: feasibility() dv_hoh

**C7** **Orbital energy Earth + Moon (+asteroid) must gain**

Formula $\Delta E = G\, M_{Sun}\, m_{sys}\, (1/(2a_0) - 1/(2a_t))$

Values $1.3271\times10^{20} \times 6.076\times10^{24} \times (1/2.992\times10^{11} - 1/8.078\times10^{11})$

Result **$1.70\times10^{33}$ J**

code: feasibility() dE_orb

### C8 Orbital angular momentum change

Formula $\Delta L = m_{sys} (\sqrt{(\mu\ a_t)} - \sqrt{(\mu\ a_0)})$

Result **$1.74\times10^{40}$ kg m²/s**

code: feasibility() dL_orb

### C9 Average power just for the orbital energy

Formula $P = \Delta E / T$

Values $1.70\times10^{33}$ J / $3.16\times10^{11}$ s

Result **$5.38\times10^{21}$ W = 30,982 × all sunlight Earth intercepts**

code: feasibility()

### C10 Escape speed from Earth + docked asteroid

Formula $v_{esc} = \sqrt{(2\ G\ (M_{Earth}+m_{ast}) / R_{Earth})}$

Values $\sqrt{(2 \times 6.6743\times10^{-11} \times 6.002\times10^{24} / 6.371\times10^{6})}$

Result **11,214 m/s**

Meaning A packet launched slower than this falls back; faster packets keep $v_\infty = \sqrt{(u^2 - v_{esc}^2)}$.

code: feasibility() vesc / launch_vinf()

### C11 Rocket equation: mass ratio if the whole asteroid is thrown

Formula $MR = m_{sys} / (m_{sys} - m_{ast})$

Values $6.0756\times10^{24} / 6.0456\times10^{24}$

Result **1.004962 (ln MR = 0.004950)**

code: feasibility()

### C12 Speed change possible at the maximum launch speed

Formula $\Delta v_{cap} = v_{\infty,max} \cdot \ln MR$, $v_{\infty,max} = \sqrt{(u_{max}^2 - v_{esc}^2)}$

Values $5.0000\times10^{6} \times 0.004950$

Result **24,750 m/s (needed 11,658 m/s)**

code: feasibility() dv_cap_mass

### C13 Reaction mass needed at the maximum launch speed

Formula $m = m_{sys} (1 - e^{-\Delta v / v_{\infty,max}})$

Result **$1.41\times10^{22}$ kg = 0.193 Moon masses**

code: feasibility() mass_needed_umax

### C14 Slowest launch speed that works with the whole asteroid (minimum energy)

Formula $v_\infty = \Delta v / \ln MR$; $u = \sqrt{(v_\infty^2 + v_{esc}^2)}$

Values $v_\infty$ = 11,658 / 0.004950 = $2.3552\times10^{6}$ m/s

Result **u = $2.3552\times10^{6}$ m/s**

code: feasibility() u_req

### C15 Minimum launch energy

Formula $E_{min} = ½\ m_{ast}\ u^2 / \eta_{launch}$

Values $0.5 \times 3.00\times10^{22} \times (2.355\times10^{6})^2 / 0.5$

Result **$1.66\times10^{35}$ J (average $5.27\times10^{23}$ W over the mission)**

code: feasibility() e_draw_min

### C16 Shortest possible transfer time (maximum packet rate and speed)

Formula $\Delta v \text{ per year} = n \cdot m_{pk} \cdot v_{\infty,max} / m_{sys}$; $t_{min} = \Delta v / (\Delta v \text{ per year})$

Values $8 \times 1.00\times10^{18} \times 5.000\times10^{6} / 6.076\times10^{24} = 6.584$ m/s per year

Result **1,771 years**

code: feasibility() years_min

### C17 Mass + time limits together

Formula $v_\infty = \max(\Delta v/\ln MR, \Delta v\, m_{sys}/(T\, n\, m_{pk}))$; $m = m_{sys}(1-e^{-\Delta v/v_\infty})$; $E = \tfrac{1}{2}\, m\, u^2/\eta$

Result **u = 2.3552×10^6 m/s, m = 3.00×10^22 kg, E = 1.66×10^35 J**

code: feasibility() combined

### C18 Launcher peak power (maximum packet rate and speed)

Formula $P_{peak} = n \cdot m_{pk} \cdot \tfrac{1}{2}\, u_{max}^2 / \eta / \text{year}$

Values $8 \times 1.00\times10^{18} \times 0.5 \times (5.00\times10^{6})^2 / 0.5 / 3.1558\times10^{7}$

Result **6.34×10^24 W**

code: feasibility() p_peak

### C19 Energy available over the mission

Formula $E_{avail} = \text{reserve} + P_{elec,min} \cdot T$

Values $3.00\times10^{35} + 1.11\times10^{25} \times 3.16\times10^{11}$

Result **3.82×10^36 J**

code: feasibility() e_avail

| Check | Result | Detail |
|---|---|---|
| mass | OK | needs launch speed 2.355e+06 m/s with ALL 3.00e+22 kg; max allowed 5.000e+06 m/s (mass-limited delta-v capacity 2.475e+04 m/s vs required 1.166e+04 m/s) |
| energy | OK | minimum energy drawn 1.664e+35 J vs available 3.817e+36 J (reserve 3.000e+35 J + collectors 1.114e+25 W x 10,000 y) |
| rate | OK | at max packet rate and speed the transfer needs >= 1,771 years (mission 10,000 y) |
| combined | OK | slowest speed meeting mass AND time limits 2.355e+06 m/s -> needs 3.000e+22 kg and 1.664e+35 J |

**Verdict:** FEASIBLE within this model (physically hypothetical scale). The realistic preset (Ceres-mass asteroid, 30 km/s launcher, Earth's own sunlight as energy) is infeasible: verdict *INFEASIBLE with these parameters (failed: mass, energy, rate, combined; mass-limited delta-v reaches 3.71e-04 of the requirement)*.

# 4. Enabler E1 - Earth launcher system

**Earth launcher system: docked asteroid + launch ring with 8 arms + arm radiators**
Throws packets of up to 1e18 kg of asteroid material at up to 5,000 km/s, 8 times per year (all 8 arms throw one packet together), with 50 % of the drawn energy ending up as packet kinetic energy; the arms reject their waste heat away from Earth; the launches do not damage Earth.

**Gap to reality:** No known structure or material; each launch carries ~1e30 J (millions of Chicxulub impacts); moving a 3e22 kg body to Earth and docking it is far beyond any technology; ~1e19 m^2 of 1,000 K radiator per mission.

**Modelled:** Momentum, energy, escape from Earth (+Moon) gravity, packet fates, finite reservoir, waste heat into the climate model. NOT modelled: the hardware, launch shock, atmosphere loss and heating of Earth (problem P6).

## Instrument E1.1 - Asteroid reservoir (docked to Earth)

**Use:** supplies the reaction mass. Earth can only change its speed by throwing mass the other way (conservation of momentum); the reservoir is finite and is used up packet by packet.

**C20 Reservoir size compared with the Moon and Earth**

| | |
|---|---|
| Formula | $m_{ast} / M_{Moon}$, $m_{ast} / M_{Earth}$ |
| Values | $3.00\times10^{22} / 7.346\times10^{22}$; $3.00\times10^{22} / 5.972\times10^{24}$ |
| Result | **0.408 Moon masses; 0.502 % of Earth** |

code: Config.asteroid_mass_kg

**C21 Surface gravity with the asteroid attached (indicative)**

| | |
|---|---|
| Formula | $g = G\,(M_{Earth} + m_{ast}) / R_{Earth}^2$ |
| Result | **9.8696 m/s² (today 9.8203 m/s², +0.50 %)** |
| Meaning | The model treats Earth + asteroid as one body (its gravity counts in every orbit calculation). |

code: indicative

**C22 Number of full packets in the reservoir**

| | |
|---|---|
| Formula | $N = m_{ast} / m_{pk,max}$ |
| Values | $3.00\times10^{22} / 1.00\times10^{18}$ |
| Result | **30,000 packets = 3,750 years at the maximum rate** |

**C23 Tested mission: reservoir used**

| | |
|---|---|
| Formula | $m_{used} / m_{ast}$ |
| Values | $2.86\times10^{22} / 3.00\times10^{22}$ |
| Result | **95.2 % in 28,648 packets** |

code: summary() mass_used_frac

## Instrument E1.2 - Launch ring with 8 arms (mass driver)

**Use:** accelerates each packet to the commanded speed u (between 20.0 and 5,000 km/s) in the commanded direction; 8 launches per year. All 8 arms throw ONE packet together (group throw), so each arm carries 1/8 of the packet and of the energy.

**C24** **Launch speed chosen by the hand-designed strategy**

| | |
|---|---|
| Formula | $v_\infty = \max(\Delta v / \ln(m_{sys}/(m_{sys}-0.95\ m_{ast})),\ \Delta v\ m_{sys} / (0.85\ T\ n\ m_{pk}))$; $u = \sqrt{(v_\infty^2+v_{esc}^2)}$ |
| Values | $\max(2.479\times10^6,\ 1.042\times10^6)$ m/s |
| Result | **$u = 2.4795\times10^6$ m/s = 2,480 km/s (0.83 % of c)** |
| Meaning | Lowest speed that uses 95 % of the asteroid and still finishes within 85 % of the mission time. |

code: heuristic_speed()

## One full packet at the strategy speed (u = 2,480 km/s)

**C25** **Kinetic energy given to packet + Earth (two-body, reduced mass)**

| | |
|---|---|
| Formula | $\mu_{red} = m_{pk}(M-m_{pk})/M$; $E_{kin} = \frac{1}{2}\ \mu_{red}\ u^2$ |
| Values | $\mu_{red} = 1.00000\times10^{18}$ kg; $0.5 \times 1.000\times10^{18} \times (2.480\times10^6)^2$ |
| Result | **$3.07\times10^{30}$ J** |

code: eject_core()

**C26** **Energy drawn from the power supply**

| | |
|---|---|
| Formula | $E_{draw} = E_{kin} / \eta_{launch}$ |
| Values | $3.07\times10^{30} / 0.5$ |
| Result | **$6.15\times10^{30}$ J per packet, $7.68\times10^{29}$ J per arm; $6.15\times10^6$ × a Chicxulub impact (~$10^{24}$ J)** |

code: eject_core()

**C27** **Packet speed after leaving Earth's and the Moon's gravity**

| | |
|---|---|
| Formula | $v_\infty = \sqrt{(u^2 - 2GM/R_{Earth} - 2GM_{Moon}/d_{Moon})}$ |
| Values | $\sqrt{((2.480\times10^6)^2 - 1.258\times10^8 - 25{,}509)}$ |
| Result | **$2.47949\times10^6$ m/s** |

code: eject_core() (fast model)

**C28** **Recoil: speed change of Earth + Moon per packet**

| | |
|---|---|
| Formula | $\Delta v_E = m_{pk}\ v_\infty / (M_{Earth}+m_{ast}+M_{Moon})$ |
| Values | $1.00\times10^{18} \times 2.479\times10^6 / 6.0756\times10^{24}$ |
| Result | **0.408 m/s (impulse $2.48\times10^{24}$ N s)** |

code: _fast_eject() / ledger LG_IMPULSE_TOTAL

**C29** **Full-throttle push per year and per decision**

| | |
|---|---|
| Formula | $\Delta v_{yr} = n \cdot \Delta v_E$ |
| Values | 8 × 0.408 |
| Result | **3.26 m/s per year, 326 m/s per 100-year decision; Δv needs 3,571 years of full thrust** |

code: policy_heuristic() dv_max

**C30** **Average launcher power at full throttle**

| | |
|---|---|
| Formula | $P = n \cdot E_{draw}$ / year |
| Values | $8 \times 6.15\times10^{30} / 3.1558\times10^7$ |
| Result | **$1.56\times10^{24}$ W ($1.95\times10^{23}$ W per arm)** |

## One full packet at the maximum speed (u = 5,000 km/s)

### C31 Kinetic energy given to packet + Earth (two-body, reduced mass)

Formula $\mu_{red} = m_{pk}(M-m_{pk})/M$; $E_{kin} = \tfrac{1}{2}\,\mu_{red}\,u^2$

Values $\mu_{red} = 1.00000\times10^{18}$ kg; $0.5 \times 1.000\times10^{18} \times (5.000\times10^{6})^2$

Result **$1.25\times10^{31}$ J**

code: eject_core()

### C32 Energy drawn from the power supply

Formula $E_{draw} = E_{kin} / \eta_{launch}$

Values $1.25\times10^{31} / 0.5$

Result **$2.50\times10^{31}$ J per packet, $3.12\times10^{30}$ J per arm; $2.50\times10^{7}$ × a Chicxulub impact (~$10^{24}$ J)**

code: eject_core()

### C33 Packet speed after leaving Earth's and the Moon's gravity

Formula $v_\infty = \sqrt{(u^2 - 2GM/R_{Earth} - 2GM_{Moon}/d_{Moon})}$

Values $\sqrt{((5.000\times10^{6})^2 - 1.258\times10^{8} - 25{,}509)}$

Result **$4.99999\times10^{6}$ m/s**

code: eject_core() (fast model)

### C34 Recoil: speed change of Earth + Moon per packet

Formula $\Delta v_E = m_{pk}\, v_\infty / (M_{Earth}+m_{ast}+M_{Moon})$

Values $1.00\times10^{18} \times 5.000\times10^{6} / 6.0756\times10^{24}$

Result **0.823 m/s (impulse $5.00\times10^{24}$ N s)**

code: _fast_eject() / ledger LG_IMPULSE_TOTAL

### C35 Full-throttle push per year and per decision

Formula $\Delta v_{yr} = n \cdot \Delta v_E$

Values $8 \times 0.823$

Result **6.58 m/s per year, 658 m/s per 100-year decision; Δv needs 1,771 years of full thrust**

code: policy_heuristic() dv_max

### C36 Average launcher power at full throttle

Formula $P = n \cdot E_{draw}$ / year

Values $8 \times 2.50\times10^{31} / 3.1558\times10^{7}$

Result **$6.34\times10^{24}$ W ($7.92\times10^{23}$ W per arm)**

### C37 Time between launches

Formula $\Delta t = 1$ year / $n$

Values 365.25 d / 8

Result **45.66 days**

code: fast_interval()

### C38 Mass per arm in a group throw

Formula $m_{arm} = m_{pk} / 8$

Values $1.00\times10^{18} / 8$

Result **$1.25\times10^{17}$ kg**

code: N_ARMS = 8

# E1.2 — Launcher-arm structural stress check

Inserted after original p.10 (C31–C38), where the report discusses maximum launch speed and the eight-arm packet split. This is a conditional static-load estimate, not a completed structural design.

## 1. Load path assumed by the equation

If a packet follows a curved guide of radius $R_c$, the guide must provide centripetal force $F=m\,v^2/R_c$. The packet applies an equal-and-opposite reaction to the launcher. The image assumes $R_c=3R_E$, eight equal load paths, and uniform axial tension over a circular cross-section of radius $r_{arm}$ in every arm.

$$\mathbf{F_{total} = m\,v^2/(3R_E)\ ;\ F_{allow} = 8\pi r_{arm}^{\,2}\sigma_{allow}}$$

Solving $F_{total}=F_{allow}$ gives $v_{max}=\sqrt{(24\pi\sigma_{allow}r_{arm}^{\,2}R_E/m)}$. With $\sigma_{allow}=86.67$ GPa, $r_{arm}=3.000\times10^{6}$ m, $m=1.00\times10^{18}$ kg and $R_E=6.371\times10^{6}$ m, this is $1.9357\times10^{7}$ m/s = 19,357 km/s (≈19,400 km/s).

| Case | Total radial load | Per arm if equally shared | Mean stress/assumed area | Use of allowable |
|---|---|---|---|---|
| **5,000 km/s operating cap** | $1.308\times10^{24}$ N | $1.635\times10^{23}$ N | 5.78 GPa | 6.67% |
| **19,357 km/s equation limit** | $1.960\times10^{25}$ N | $2.451\times10^{24}$ N | 86.67 GPa | 100% |

At 5,000 km/s, speed is 25.8% of the equation limit and the stress is 6.67% of the stated allowable because stress scales as $v^2$. If 86.67 GPa already includes safety factor 1.5, the implied ultimate strength is about 130 GPa; the nominal margin to that assumed ultimate is 22.5×. This is only for the stated idealized load.

## Geometry and real arm stresses

The expression uses $\pi r_{arm}^{\,2}$ as the cross-sectional area of each arm. If 3,000 km means arm reach/length rather than cross-section radius, that substitution is invalid. The path curvature, attachment points, load sharing and force transfer must be specified. Real arms may carry tension, compression, shear and bending; the calculation omits arm self-weight, buckling, joints, stress concentrations, launch transients, fatigue and temperature-dependent strength. Use actual area A and path radius $R_c$: $v_{max}=\sqrt{(N\sigma_{allow}A\,R_c/m)}$, only if equal sharing and this load model are justified.

# E1.2 — Thermal loading and combined arm stress

The 50% launcher efficiency provides a heat-energy estimate; it does not determine arm temperature or thermal stress. The heat flow, launch pulse, and material properties are not specified in the original report.

## 2. Heat implied at the 5,000 km/s setting

| Quantity | Value | Basis |
|---|---|---|
| **Packet kinetic energy** | $1.25\times10^{31}$ J | ½mv² for $m=10^{18}$ kg, $v=5\times10^{6}$ m/s |
| **Energy drawn at η=0.5** | $2.50\times10^{31}$ J | $E_{kin}/\eta$ |
| **Waste heat per packet/event** | $1.25\times10^{31}$ J | $E_{draw}-E_{kin}$, assuming loss becomes heat |
| **If divided across 8 arms** | $1.56\times10^{30}$ J per arm/event | Equal split assumption only |
| **At 8 events/year** | $3.17\times10^{24}$ W total average; $3.96\times10^{23}$ W/arm average | Average over the year, not pulse power/heat flux |

This is a system-level energy ledger. Some inefficiency may heat power electronics, rectennas, launcher conductors or radiators rather than the arm itself. If fraction $f_{arm}$ of the per-arm event heat is deposited locally over pulse time $t_p$, a first estimate of pulse heating power is $f_{arm}(1.56\times10^{30}\ \text{J})/t_p$. Neither $f_{arm}$ nor $t_p$ is provided.

## 3. Thermal stress and the combined design check

For a uniformly heated member whose expansion is fully restrained, a one-dimensional estimate is $\sigma_{th}=E\alpha\Delta T$. A freely expanding member under uniform heating has little axial thermal stress; gradients, constraints, joints and rapid heating/cooling create local stress, bending and fatigue. Combine the mechanical, thermal and dynamic stress tensors at the most-loaded location/time; check an appropriate equivalent-stress/failure criterion against allowable strength at operating temperature. Also assess thermal-cycle fatigue and creep.

| Illustration only (not the specified arm material) | Fully restrained thermal stress |
|---|---|
| $E=200$ GPa; $\alpha=12\times10^{-6}$/K | 2.4 MPa per K |
| $\Delta T=100$ K / 500 K | 0.24 GPa / 1.20 GPa |

The reported 3,600-year run has 28,648 packets, so the launcher would face roughly that many load/thermal cycles; 8 events/year over a full 10,000-year mission would be 80,000 cycles. A single static allowable does not establish fatigue life.

## Inputs still needed

Named material and strength-vs-temperature curve; actual load-bearing area and arm geometry; support/track curvature; force-time profile and load sharing; E, α, thermal conductivity, heat capacity and emissivity; component-level heat deposition, pulse duration, cooling and radiator geometry; temperature field; cyclic fatigue/creep and multiaxial stress analysis. Until then, 5,000 km/s is below the equation's static cap only conditionally—not demonstrated safe.

**C39** **Tested mission: total impulse and speed change delivered**

Formula $\Delta v = J_{total} / M_{Earth+Moon+ast}$

Values $7.08\times10^{28}$ N s / $6.0756\times10^{24}$ kg

Result **11,657 m/s (required by the spiral estimate: 11,658 m/s)**

Meaning Compare with the slow-spiral estimate C5: the hand-designed strategy pushes almost purely along Earth's motion, so the delivered speed change is close to the ideal value.

code: ledger LG_IMPULSE_TOTAL

**C40** **Tested mission: packet kinetic energy and launcher draw**

Formula $\eta = E_{kin} / E_{draw}$

Values $8.78\times10^{34}$ / $1.76\times10^{35}$

Result **0.5000 (= launch_efficiency 0.5)**

code: ledger LG_LAUNCH_KE, LG_E_DRAWN

**C41** **Tested mission: average launcher power**

Formula $P = E_{draw} / t$

Values $1.76\times10^{35}$ / (3,600 y × $3.1558\times10^{7}$ s)

Result **$1.55\times10^{24}$ W**

**Not modelled** (assumed by E1): the arms' structure, materials and length, the acceleration of the packets inside the arms, and the effect of each launch on Earth itself (shock, atmosphere loss, heating). Open problem P6.

## Instrument E1.3 - Arm power conditioning and energy reserve

**Use:** the arms take electricity from the rectennas (E2) first; if a launch needs more than the beams deliver in that interval, the rest comes from a finite reserve. Unused beam energy is not sent (the beams are throttled to demand); nothing is created from nothing.

**C42** **Reserve size compared with the minimum mission energy**

Formula $reserve / E_{min}$

Values $3.00\times10^{35}$ / $1.66\times10^{35}$

Result **1.803**

code: Config.energy_budget_j

**C43** **Energy offered by the beams in one decision interval (Earth at 1 AU)**

Formula $E_{beam} = P_{elec} \cdot \Delta t_{decision}$

Values $1.29\times10^{25}$ W × 100 y

Result **$4.07\times10^{34}$ J**

code: _PowerSupply._with_supply()

**C44** **Tested mission: where the launch energy came from**

Formula $E_{draw} = E_{beams} + E_{reserve}$

Values $1.76\times10^{35}$ + 0

Result **$1.76\times10^{35}$ J; beams used 12.8 % of what they could offer**

code: summary() energy_from_collectors_J

## Instrument E1.4 - Directional arm radiators

**Use:** the launcher loses (1−η) = 50 % of the energy it draws as heat, and the rectennas lose ($1-\eta_r$) = 15 % of the microwaves they catch. The radiators send this heat to space, away from Earth; only the share

arm_heat_to_earth_frac = $1.00\times10^{-9}$ reaches Earth.

**C45** **Tested mission: arm waste heat**

Formula $Q = E_{\text{rectenna heat}} + E_{\text{launcher heat}}$

Values $3.10\times10^{34} + 8.78\times10^{34}$ J

Result **$1.19\times10^{35}$ J = $1.05\times10^{24}$ W on average = $6.03\times10^{6}$ × the sunlight Earth intercepts**

code: settle_interval()

**C46** **Radiator area needed at 1,000 K (indicative, black radiator, one side)**

Formula $A = Q / (\sigma\ T\&sup4;)$

Values $1.05\times10^{24}$ / ($5.670\times10^{-8}$ × 1000&sup4;)

Result **$1.84\times10^{19}$ m² in total, $2.31\times10^{18}$ m² per arm = 18,077 × Earth's disc**

code: summary() radiator_area_1000K_m2

**C47** **Heat that still reaches Earth**

Formula $q_{\text{Earth}} = f_{\text{arm}} \cdot Q + f_{\text{spill}} \cdot E_{\text{spill}}$

Values $1.00\times10^{-9} \times 1.19\times10^{35} + 1.00\times10^{-9} \times 5.48\times10^{33}$

Result **$1.24\times10^{26}$ J = $1.09\times10^{15}$ W = 0.00630 × Earth's sunlight**

code: _climate_step() q_w

Without the directional radiators a geometric leak of ~8 % would put $8.37\times10^{22}$ W ($4.82\times10^{5}$ × Earth's sunlight) on Earth: the oceans would boil. This is why the radiators are part of enabler E1.

## Mars safety limits (control software, not an enabler)

Earth's growing orbit crosses Mars's orbit. The safety shield forecasts the Earth-Mars distance and overrides the agent (coast or full thrust) when a pass would come too close. The safe distance is the largest of three limits:

**C48** **Tide limit: Mars's tide on Earth ≤ 1 % of the Moon's**

Formula $d_{\text{tide}} = d_{\text{Moon}} (GM_{\text{Mars}} / (GM_{\text{Moon}} \cdot k_{\text{tide}}))^{1/3}$

Values $3.844\times10^{8} \times (4.283\times10^{13} / (4.903\times10^{12} \times 0.01))^{1/3}$

Result **$3.67\times10^{6}$ km**

code: mars_safety_limits()

**C49** **Moon-orbit limit: Mars's pull on the Moon's orbit ≤ 0.1 % of Earth's**

Formula $d_{\text{moon}} = d_{\text{Moon}} (2\ GM_{\text{Mars}} / (GM_{\text{Earth}}\ k_{\text{moon}}))^{1/3}$

Values $k_{\text{moon}} = 0.001$

Result **$2.30\times10^{6}$ km**

code: mars_safety_limits()

**C50** **Safe distance and shield caution distance**

Formula $d_{\text{safe}} = \max(d_{\text{tide}}, d_{\text{moon}}, d_{\text{min}})$; $d_{\text{caution}} = k_c\ d_{\text{safe}}$

Values $\max(3.67\times10^{6}, 2.30\times10^{6}, 2.00\times10^{6})$ km; $k_c = 2.0$

Result **$d_{\text{safe}} = 3.67\times10^{6}$ km = 0.0246 AU (governed by: tide on Earth); caution 0.0491 AU**

code: mars_safety_limits()

Tested mission: closest Earth-Mars pass 0.0494 AU (≥ 0.0246 AU: safe), largest tide ratio 0.00123, 176 shield overrides. Mars's own orbit changed by Δa = 0.00606 AU, Δe = $8.75\times10^{-4}$.

# 5. Enabler E2 - Solar power belt with microwave power beaming

**Solar power belt with microwave power beaming (collectors + transmitters + arm rectennas)**
5 million thin-film collectors at 0.5 AU intercept 30 % of the Sun's output, turn it into electricity and beam it as 5.8 GHz microwaves (30 km phased arrays) to 1,000 km rectennas on the 8 arms; they hold their slots by sail tilt, keep a clear sunlight lane for Earth and Mars, keep beam spill off Earth and are built up / replaced by factories using Mercury material.

**Gap to reality:** A Dyson-swarm-scale project (~1e13 x all solar panels ever built); record power beaming is ~kW over km, here ~1e24 W over 0.5-3 AU with pointing to ~1e-12 rad; beam intensity at the rectennas ~2e8 x sunlight.

**Modelled:** Panel temperature and efficiency, degradation and replacement, build-up, manufacturing energy and mass, Goubau diffraction capture, Sun blocking, every loss in the energy ledger, clear lanes, the orbits of 12+12 followed collectors with / without station keeping (beam recoil, planets, light pressure).

## Instrument E2.1 - Thin-film solar collectors

**Use:** intercept sunlight and turn it into electricity. The belt is a band around the Sun at 0.5 AU that covers the fraction f = 0.3 of a full sphere, so it intercepts exactly that fraction of all sunlight.

### C51 Belt area

Formula $A = f \cdot 4\pi R^2$

Values $0.3 \times 4\pi \times (7.480 \times 10^{10})^2$

Result **$2.11 \times 10^{22}$ m² = $4.14 \times 10^{7}$ × Earth's surface**

code: collector_design()

### C52 Band half-width (latitude seen from the Sun)

Formula $\beta_{band} = \arcsin f$

Values $\arcsin 0.3$

Result **17.46°**

code: collector_design()

### C53 Size of one collector

Formula $A_1 = A / N$; side = $\sqrt{A_1}$

Values $2.11 \times 10^{22} / 5.00 \times 10^{6}$

Result **$4.22 \times 10^{15}$ m², about 64,950 km × 64,950 km**

code: collector_design()

### C54 Mass of the belt and of one collector

Formula $m = A \cdot \sigma_{areal}$

Values $2.11 \times 10^{22} \times 0.1$ kg/m²

Result **$2.11 \times 10^{21}$ kg (0.0287 Moon masses); one collector $4.22 \times 10^{14}$ kg**

code: collector_design()

### C55 Sunlight at the belt

Formula $S = S_{1AU} / r^2$

Values $1361.0 / 0.5^2$

Result **5,444.0 W/m²**

code: collector_operating()

**C56** **Sunlight intercepted (whole belt built)**

Formula $L_{cap} = f \cdot L_{Sun}$

Values $0.3 \times 3.828 \times 10^{26}$

Result **$1.15 \times 10^{26}$ W**

code: collector_power()

**C57** **Panel temperature (energy balance, both faces radiate)**

Formula $T = [(a\,S - \eta(T)\,S) / (2\,\varepsilon\,\sigma)]^{1/4}$, $\eta(T) = \eta_{25°C}(1 - k\,(T - 298.15\ K))$

Values $a = 0.9$, $\varepsilon = 0.85$, $\eta_{25°C} = 0.25$, $k = 0.002/K$ (solved by iteration)

Result **T = 450.0 K, η(T) = 17.41 % (×0.696); idle panel (no electricity) 474.8 K**

code: collector_operating()

**C58** **Ageing of the fleet (panels replaced at end of life)**

Formula $f_{age} = 1 - d \cdot L_{life} / 2$

Values $1 - 0.002 \times 50 / 2$

Result **0.950**

code: collector_operating()

**C59** **Operating solar-cell efficiency**

Formula $\eta_{op} = \eta(T) \cdot f_{age}$

Values $0.1741 \times 0.950$

Result **16.54 %**

code: collector_operating()

**C60** **Panel electricity (whole belt, clear lanes subtracted)**

Formula $P_{pv} = f\,L_{Sun}\,(1 - lanes)\,\eta_{op}$

Values $1.15 \times 10^{26} \times (1 - 5.11 \times 10^{-5}) \times 0.1654$

Result **$1.90 \times 10^{25}$ W; one collector $3.80 \times 10^{18}$ W**

code: collector_power()

**C61** **Light-pressure ratio of a collector (push of sunlight / pull of the Sun)**

Formula $\beta = L_{Sun} / (4\pi\,c\,GM_{Sun}\,\sigma_{areal})$

Values $3.828 \times 10^{26} / (4\pi \times 2.998 \times 10^{8} \times 1.327 \times 10^{20} \times 0.1)$

Result **0.00766 (0.77 %)**

code: collector_design()

**C62** **Orbital period of a collector**

Formula $P = r^{3/2} / \sqrt{(1 - \beta)}$ years

Values $0.5^{1.5} / \sqrt{(1 - 0.0077)}$

Result **0.3549 years = 129.6 days**

code: collector_design()

**C63** **Gravity of the whole belt on Earth (vs the Sun's pull)**

Formula $a = \Sigma\,G\,m_i\,d_i/|d_i|^3$ over 6,480 belt points

Result **$1.28 \times 10^{-9}$ of the Sun's pull at 1 AU, $1.09 \times 10^{-9}$ at 2.7 AU (negligible)**

code: belt_pull_on_earth()

## Instrument E2.2 - Microwave phased-array transmitters

**Use:** each collector turns its electricity into a 5.8 GHz beam with a 30 km phased array and points it at the arms. Efficiency 80 %; the beams are throttled to what the launcher needs.

### C64 Wavelength

Formula $\lambda = c / f$

Values $2.99792\times10^{8} / 5.80\times10^{9}$

Result **5.169 cm**

code: beam_capture_efficiency()

### C65 Microwave power sent (whole belt, full output)

Formula $P_{mw} = P_{pv}\,\eta_{mw}$

Values $1.90\times10^{25} \times 0.8$

Result **$1.52\times10^{25}$ W; per transmitter $3.04\times10^{18}$ W**

code: collector_power()

### C66 Intensity on a transmitter aperture at full output (indicative)

Formula $I = (P_{mw}/N) / (\pi\,(D_t/2)^2)$

Values $3.04\times10^{18} / 7.07\times10^{8}$ m²

Result **$4.30\times10^{9}$ W/m² = $3.16\times10^{6}$ × sunlight at Earth**

code: indicative

### C67 Beam capture at 0.50 AU (Goubau relation)

Formula $\tau = D_t\,D_r / (\lambda\,d);\ \eta_{cap} = 1 - e^{-\tau^2}$

Values $\tau = 30{,}000 \times 1.00\times10^{6} / (0.05169 \times 7.480\times10^{10}) = 7.759$

Result **$\eta_{cap}$ = 100.00 %**

code: beam_capture_efficiency()

### C68 Central beam spot diameter at 0.50 AU (indicative, Airy)

Formula $D_{spot} \approx 2.44\,\lambda\,d / D_t$

Values $2.44 \times 0.05169 \times 7.480\times10^{10} / 30{,}000$

Result **314 km (rectenna 1000 km)**

code: indicative

### C69 Beam capture at 1.50 AU (Goubau relation)

Formula $\tau = D_t\,D_r / (\lambda\,d);\ \eta_{cap} = 1 - e^{-\tau^2}$

Values $\tau = 30{,}000 \times 1.00\times10^{6} / (0.05169 \times 2.244\times10^{11}) = 2.587$

Result **$\eta_{cap}$ = 99.88 %**

code: beam_capture_efficiency()

### C70 Central beam spot diameter at 1.50 AU (indicative, Airy)

Formula $D_{spot} \approx 2.44\,\lambda\,d / D_t$

Values $2.44 \times 0.05169 \times 2.244\times10^{11} / 30{,}000$

Result **943 km (rectenna 1000 km)**

code: indicative

**C71** **Average capture over the whole belt, Earth at 1 AU / at the target**

Formula $\bar{\eta}_{cap}$ = mean over 6,480 belt points (beams within 2 $R_{Sun}$ of the Sun blocked)

Values blocked: 0.1 % / 0.1 % of the belt

Result **99.88 % / 86.30 %**

code: collector_power()

**C72** **Pointing accuracy needed (indicative)**

Formula $\delta\theta \approx (D_r/2) / d_{max}$

Values $5.00\times10^{5}$ m / $4.787\times10^{11}$ m

Result **$1.04\times10^{-6}$ rad**

code: indicative

**C73** **Tested mission: transmitter waste heat and beam spill**

Formula $Q_{tx} = E_{el}(1-\eta_{mw})$; $E_{spill} = E_{mw}(1-\eta_{cap})$

Values $2.65\times10^{35}$ × 0.20; $2.12\times10^{35}$ − $2.07\times10^{35}$

Result **$Q_{tx} = 5.30\times10^{34}$ J; spill $5.48\times10^{33}$ J (2.58 %), of which $1.00\times10^{-9}$ reaches Earth**

code: settle_interval()

## Instrument E2.3 - Rectennas on the 8 arm tips

**Use:** receive the microwaves (only the arms receive power) and turn them back into electricity with 85 % efficiency. Diameter 1000 km each.

**C74** **Area of one rectenna**

Formula $A_r = \pi (D_r/2)^2$

Values $\pi \times (5.00\times10^{5})^2$

Result **$7.85\times10^{11}$ m² (0.0493 × Earth's disc for all 8)**

code: collector_design()

**C75** **Microwaves arriving (full output), Earth at 1 AU / at the target**

Formula $P_{in} = P_{mw}\,\bar{\eta}_{cap}$

Values $1.52\times10^{25}$ × 0.9988; $1.52\times10^{25}$ × 0.8630

Result **$1.52\times10^{25}$ W / $1.31\times10^{25}$ W**

code: collector_power()

**C76** **Electricity delivered to the arms (full output)**

Formula $P_{elec} = P_{in}\,\eta_r$

Values $1.52\times10^{25}$ × 0.85

Result **$1.29\times10^{25}$ W at 1 AU, $1.11\times10^{25}$ W at 2.7 AU; at year 0 (only 30 % built) $3.87\times10^{24}$ W**

code: collector_power()

**C77** **Supply vs the launcher's peak demand**

Formula $P_{elec,min} / P_{peak}$

Values $1.11\times10^{25}$ / $6.34\times10^{24}$

Result **1.76 ×**

code: feasibility()

**C78** **Beam intensity on a rectenna at full output (Earth at 1 AU)**

Formula $I = P_{in} / (8\ A_r)$

Values $1.52\times10^{25}$ / ($8 \times 7.85\times10^{11}$)

Result **$2.42\times10^{12}$ W/m² = $1.77\times10^{9}$ × sunlight**

code: collector_design()

**C79** **Tested mission: average and peak beam power received**

Formula $\bar{P} = E_{captured} / t$

Values $2.07\times10^{35}$ J / (3,600 y)

Result **mean $1.82\times10^{24}$ W ($2.27\times10^{23}$ W per arm, $2.89\times10^{11}$ W/m²); peak $1.83\times10^{24}$ W**

code: summary() beam_mean_W

**C80** **Tested mission: rectenna heat**

Formula $Q_r = E_{captured}\ (1 - \eta_r)$

Values $2.07\times10^{35} \times 0.15$

Result **$3.10\times10^{34}$ J (removed by the E1 radiators)**

code: settle_interval()

**C81** **Tested mission: push of the absorbed microwaves on Earth (indicative side effect)**

Formula $\Delta v = E_{in} / (c\ M)$

Result **114 m/s (counted in the report, not applied)**

code: summary() beam_push_dv_ms

## Instrument E2.4 - Sail-tilt station keeping

**Use:** each collector tilts slightly so the sunlight push corrects its orbit (no propellant). It may use up to collector_sk_authority = 0.5 of its light push. 12 collectors are simulated with it and the same 12 without it (RK4, half-day steps) along Earth's and Mars's simulated motion.

**C82** **Sun's pull at the belt**

Formula $g = GM_{Sun} / R^2$

Values $1.3271\times10^{20}$ / $(7.480\times10^{10})^2$

Result **0.0237 m/s²**

code: BeltKeeper

**C83** **Largest control acceleration**

Formula $a_{ctrl} = k_{sk}\ \beta\ GM_{Sun} / R^2$

Values 0.5 × 0.00766 × 0.0237

Result **$9.08\times10^{-5}$ m/s² = 0.00383 of the Sun's pull**

code: BeltKeeper.auth

**C84** **Controller speed (PD gains)**

Formula $\omega_c = 4\ n,\ n = \sqrt{(GM_{Sun}(1-\beta)/R^3)}$

Values $n = 5.61\times10^{-7}$ rad/s

Result **$\omega_c = 2.24\times10^{-6}$ rad/s (time constant 5.2 days)**

code: BeltKeeper.wc

**C85** **Beam recoil on a collector (full output)**

Formula $a_{rec} = (P_{mw}/N) / (c\ m_1)$

Values $3.04\times10^{18} / (2.998\times10^{8} \times 4.22\times10^{14})$

Result **$2.40\times10^{-5}$ m/s² = 0.00101 of the Sun's pull**

code: BeltKeeper.advance() a_rec

**C86** **Pull of a neighbouring collector**

Formula $a = G\ m_1 / s^2$

Values $6.6743\times10^{-11} \times 4.22\times10^{14} / (6.495\times10^{7})^2$

Result **$2.81\times10^{-10}$ of the Sun's pull**

code: BeltKeeper.neighbour_ratio

| Tested mission (3,600 years) | WITH station keeping | WITHOUT |
|---|---|---|
| largest distance from the slot | 723 km | $1.46\times10^{8}$ km |
| ... in collector spacings (64,950 km) | 0.0111 | 2,244 |
| orbit size change \|Δa\| | 1,487 km | 21,309 km |
| largest eccentricity | $1.21\times10^{-5}$ | 0.127 |
| control effort | 96.8 m/s per year | 0 |

Largest disturbances (fraction of the Sun's pull): beam recoil $1.37\times10^{-4}$, planets $2.48\times10^{-6}$, neighbours $2.81\times10^{-10}$; the controller can correct up to 0.00383.

## Instrument E2.5 - Clear sunlight lane

**Use:** the collectors leave a moving gap on the line Sun-Earth (and Sun-Mars) so that Earth sees the whole solar disc.

**C87** **Share of the belt that must stay empty for Earth**

Formula $\text{lane} = \rho^2 / (4\ R^2\ f),\ \rho = R_{Sun}(r-R)/r + R_{Earth}\ R/r$

Values r = 1 AU

Result **$1.84\times10^{-5}$ (Mars: $3.27\times10^{-5}$)**

code: lane_fraction()

**C88** **Power given up for the lanes**

Formula $\Delta P = L_{cap} \times \text{lanes} \times \eta_{op}$

Values $1.15\times10^{26} \times 5.11\times10^{-5} \times 0.1654$

Result **$9.70\times10^{20}$ W of electricity**

code: collector_power()

## Instrument E2.6 - Collector factories (Mercury material)

**Use:** 30 % of the belt exists at year 0; factories finish it in 400 years and replace panels after 50 years (95 % recycled). Manufacturing energy comes from the collectors' own output.

**C89** **Built share over time**

Formula $b(t) = b_0 + (1 - b_0)\ t / T_{build}\ (\max 1)$

Values $b_0 = 0.3$, $T_{build}$ = 400 y

Result **b(100 y) = 47.5 %, b(200 y) = 65.0 %, complete at 400 y**

code: collector_built()

**C90** **Replacement mass per year (whole belt)**

Formula $\dot{m} = m_{belt} / L_{life}$

Values $2.11\times10^{21}$ / 50

Result **$4.22\times10^{19}$ kg per year**

code: collector_design()

**C91** **Fresh material over the mission**

Formula $m_{fresh} = m_{belt} + \dot{m}\,(1 - recycle)\,T$

Values $2.11\times10^{21}$ + $4.22\times10^{19}$ × 0.05 × 10,000

Result **$2.32\times10^{22}$ kg = 7.0 % of Mercury's mass**

code: collector_design()

**C92** **Manufacturing energy over the mission**

Formula $E = e_{emb}\,(m_{belt}(1 - b_0) + \dot{m}\,T)$

Values $e_{emb} = 1.00\times10^{8}$ J/kg

Result **$4.23\times10^{31}$ J = 0.0706 years of the belt's full electricity**

code: collector_design()

Tested mission: manufacturing energy charged $1.47\times10^{31}$ J; panels built $1.47\times10^{23}$ kg, fresh material $8.77\times10^{21}$ kg.

# 6. Enabler E3 - Space mirror ring (soletta) and the climate

**Space mirror ring around Earth (soletta)**
1.6e15 m^2 of aluminium-film mirrors (8e12 kg) in a terminator orbit 20 Earth radii out reflect extra sunlight onto Earth: up to 1.10 x today's sunlight at 2.7 AU. The agent sets how much (mirrors not needed are turned so their light passes Earth).

**Gap to reality:** Physically possible (only reflection, no new physics, no energy needed). Demonstrated small: Znamya-2 (1993) 20 m mirror lit a 5 km spot on Earth; solar sails IKAROS (2010, ~200 m^2). Needed: ~1e13 x that area; film made from asteroid aluminium; formation flying and pointing of a million 40 km mirrors.

**Modelled:** Capacity (1/r^2), light sent, light turned away, mirror heat (energy ledger), mass; the climate effect. Not the mirrors' orbits, light pressure, pointing errors or the light pattern on Earth.

## Instrument E3.1 - Mirror film panels

**Use:** reflect sunlight onto Earth. Aluminium film, 5 g/m², reflectivity 90 %. Mirrors that are not needed are turned so their light passes Earth.

**C93** **Mirror mass**

Formula $m = A\ \sigma_{areal}$

Values $1.60\times10^{15} \times 0.005$

Result **$8.00\times10^{12}$ kg**

code: mirror_design()

**C94** **Area compared with Earth's disc; equivalent single disc**

Formula $A / (\pi\ R_{Earth}^2)$; $r = \sqrt{(A/\pi)}$

Values $1.60\times10^{15} / 1.28\times10^{14}$

Result **12.55 ×; one disc of radius 22,568 km, or 1,273,240 mirrors of 40 km**

code: mirror_design()

**C95** **Area facing the Sun (45° tilt)**

Formula $A_{eff} = A \cos 45°$

Values $1.60\times10^{15} \times 0.7071$

Result **$1.13\times10^{15}$ m²**

code: mirror_design()

**C96** **Largest light the mirrors can send to Earth, Earth at 1 AU**

Formula $P_{cap} = A_{eff}\ \rho\ S_{1AU} / r^2$

Values $1.13\times10^{15} \times 0.9 \times 1361.0 / 1^2$

Result **$1.39\times10^{18}$ W = 7.985 × the sunlight Earth intercepts ($1.74\times10^{17}$ W)**

code: mirror_design()

**C97** **... Earth at the target (2.7 AU)**

Formula $P_{cap} = A_{eff}\ \rho\ S_{1AU} / a_t^2$

Values $1.13\times10^{15} \times 0.9 \times 1361.0 / 2.7^2$

Result **$1.90\times10^{17}$ W = 1.095 × today's sunlight on Earth**

code: mirror_design()

**C98** **Mirror light needed at 2.7 AU to restore today's sunlight**

Formula $P_{need} = (1 - 1/a_t^2)\, S_{1AU}\, \pi\, R_{Earth}^2$

Values $(1 - 0.1372) \times 1.74\times10^{17}$

Result **$1.50\times10^{17}$ W = 0.788 of the mirror capacity (margin 26.9 %)**

code: climate_integrate()

**C99** **Heat absorbed by the mirrors**

Formula $Q = (1 - \rho) \times$ sunlight on the mirrors

Values $1 - 0.9$

Result **10 % of the sunlight they catch (radiated to space)**

code: settle_interval()

**C100** **Light pressure on a mirror at 2.7 AU (indicative, NOT simulated)**

Formula $a = 2\, S \cos^2 45° / (c\, \sigma_{areal})$

Values $2 \times 186.7 \times 0.5 / (2.998\times10^8 \times 0.005)$

Result **$1.25\times10^{-4}$ m/s² = 0.51 % of Earth's gravity at the mirror orbit**

Meaning Small but not negligible: the mirrors need active orbit and attitude control (part of E3, not simulated).

code: indicative

## Instrument E3.2 - Terminator orbit and pointing

**Use:** the ring orbits Earth at 20 Earth radii in a plane at right angles to the sunlight, so each mirror always sees the Sun at 90° from Earth and is tilted 45°. The agent's action a4 sets how much light is aimed at Earth.

**C101** **Orbit radius and period**

Formula $r = k\, R_{Earth}$; $P = 2\pi \sqrt{(r^3/(G\, M_{Earth}))}$

Values $r = 20 \times 6.371\times10^6 = 1.274\times10^8$ m

Result **P = 5.24 days**

code: mirror_design()

**C102** **Size of the light spot on Earth from one mirror (Sun's angular size)**

Formula $D = \theta_{Sun}\, r$, $\theta_{Sun} = 2\, R_{Sun} / (a\ AU)$

Values $\theta_{Sun}$ at 2.7 AU = 0.00344 rad

Result **439 km at 2.7 AU (1,185 km at 1 AU)**

code: mirror_design()

**C103** **Pointing accuracy to keep the spot on Earth (indicative)**

Formula $\delta\theta \approx R_{Earth} / r$

Values $6.371\times10^6 / 1.274\times10^8$

Result **0.0500 rad = 2.86° (a mirror turns by half that)**

code: indicative

## Climate model (what the mirrors must achieve)

Two-box energy balance: a surface layer (atmosphere + ocean mixed layer) exchanging heat with the deep ocean; Earth's reflectivity rises from 0.30 to 0.60 as it freezes (snowball risk).

**C104** **Effective emissivity (fixed greenhouse, 288 K today)**

Formula $\varepsilon_{eff} = S_{1AU}(1-\alpha_{warm})/4 / (\sigma$ 288&sup4;)

Values 1361.0 × 0.70 / 4 / (5.670×10$^{-8}$ × 288&sup4;)

Result **0.6105**

code: CLIM_EPS

**C105** **Surface temperature equation**

Formula $C_s\, dT_s/dt = S(1-\alpha(T_s))/4 + q_w/(4\pi R^2) - \varepsilon\sigma T_s$&sup4; $- \gamma(T_s-T_d)$; $C_d\, dT_d/dt = \gamma(T_s-T_d)$

Values $C_s$ = 8.0 W yr m$^{-2}$K$^{-1}$, $C_d$ = 100.0, γ = 0.7 W m$^{-2}$K$^{-1}$, step ≤ 0.0625 y

Result **surface response time ≈ $C_s/(4\varepsilon\sigma T^3 + \gamma)$ = 2.0 years**

code: climate_integrate()

**C106** **Equilibrium at 2.7 AU WITHOUT mirrors**

Formula solve $S(1-\alpha(T))/4 = \varepsilon\sigma$T&sup4;, $S = S_{1AU}/a_t^2$

Values S = 186.7 W/m² (0.1372 × today)

Result **152.4 K (-121 °C): snowball (frozen to the equator)**

code: climate_equilibrium()

**C107** **Equilibrium at 2.7 AU with ALL mirror light on Earth**

Formula $S = S_{1AU}(1/a_t^2 + P_{cap}/(S_{1AU}\pi R^2))$

Values 0.1372 + 1.0954 = 1.2325 × today

Result **303.5 K — the agent aims less light (a set-point) to stay at 288 K**

code: climate_equilibrium()

**C108** **Life indicators at the target temperature**

Formula $T(lat) = \bar{T} + \Delta T/3 - \Delta T \sin^2(lat)$, $\Delta T$ = 45 K; ice where T < 263 K; growth for 278-313 K

Values $\bar{T}$ = 288 K

Result **equator 303 K, pole 258 K, ice line 70.5°, growing area 74.5 %: temperate, like today**

code: life_indicators()

**C109** **Tested mission: mirror light**

Formula sunlight on mirrors = to Earth + turned past Earth + heat

Values 8.17×10$^{28}$ = 1.04×10$^{28}$ + 6.32×10$^{28}$ + 8.17×10$^{27}$ J

Result **surface 288.3 K (largest deviation 2.6 K, 0 years outside the livable band); without mirrors it would be 157 K**

code: settle_interval(), summary()

# 7. Energy and mass ledger of the tested mission

| Flow (J) | Hand-designed mission, high fidelity |
|---|---|
| IN: sunlight on the collectors | $1.25\times10^{37}$ |
| IN: reserve used | 0 |
| IN: sunlight on the mirrors | $8.17\times10^{28}$ |
| reflected by the panels | $1.25\times10^{36}$ |
| collector heat | $1.10\times10^{37}$ |
| manufacturing (new + replaced panels) | $1.47\times10^{31}$ |
| panel electricity used | $2.65\times10^{35}$ |
| transmitter heat | $5.30\times10^{34}$ |
| microwaves sent | $2.12\times10^{35}$ |
| beam spill (missed the rectennas) | $5.48\times10^{33}$ |
| captured by the rectennas | $2.07\times10^{35}$ |
| rectenna heat | $3.10\times10^{34}$ |
| arm electricity | $1.76\times10^{35}$ |
| launcher draw | $1.76\times10^{35}$ |
| packet kinetic energy | $8.78\times10^{34}$ |
| launcher heat | $8.78\times10^{34}$ |
| mirror light onto Earth | $1.04\times10^{28}$ |
| mirror light turned past Earth | $6.32\times10^{28}$ |
| mirror heat | $8.17\times10^{27}$ |
| waste heat reaching Earth | $1.24\times10^{26}$ |

**C110** **Energy balance**

Formula: IN = reflected + collector heat + manufacturing + transmitter heat + spill + rectenna heat + packet KE + launcher heat + mirror flows

Values: $1.25385\times10^{37}$ vs $1.25385\times10^{37}$ J

Result: **relative residual $1.88\times10^{-16}$ (round-off level: no energy created or lost)**

code: settle_interval() residual_rel

**C111** **Efficiency of the beaming chain (panel electricity to arm electricity)**

Formula: $\eta = \eta_{mw}\,\bar{\eta}_{cap}\,\eta_{r}$

Values: $1.76\times10^{35}$ / $2.65\times10^{35}$

Result: **66.2 %**

code: settle_interval()

**C112** **Sunlight-to-packet efficiency**

Formula: $E_{\text{packet KE}}$ / $E_{\text{sun on used panels}}$

Values: $8.78\times10^{34}$ / ($2.65\times10^{35}$ / 0.1654)

Result: **5.48 %**

| Mechanical conservation (N-body, whole mission) | Relative residual |
|---|---|
| energy (system vs launches, escapes, impacts) | $1.53\times10^{-6}$ |
| momentum | $1.96\times10^{-13}$ |
| angular momentum | $4.00\times10^{-13}$ |
| mass | $1.42\times10^{-16}$ |

Mass: $2.86\times10^{22}$ kg ejected in 28,648 packets; $2.86\times10^{22}$ kg left the Solar System; $1.00\times10^{18}$ kg hit the Moon; 0 kg fell back to Earth; 0 kg hit Mars.

# 8. Tested results and the control system

| High-fidelity mission (seed 2001) | No ejection | Hand-designed heuristic | Best PPO model |
|---|---|---|---|
| result | deadline | success | success |
| final a / e | 1.0000 AU / 0.0159 | 2.7000 AU / 0.0081 | 2.6870 AU / 0.0085 |
| simulated years | 10,000 | 3,600 | 7,100 |
| asteroid used | 0.0 % | 95.2 % | 96.5 % |
| energy drawn | 0 J | $1.76\times10^{35}$ J | $1.80\times10^{35}$ J |
| surface temperature | 288.2 K | 288.3 K | 286.8 K |
| closest Mars pass | 0.3721 AU | 0.0494 AU | 0.0413 AU |
| reward (scoring only, not physics) | -54.1 | 214.9 | 193.0 |

The hand-designed strategy reaches the target; the trained PPO checkpoint also reaches it in this test but it is an early lucky policy (later training collapsed). Neither is claimed to be optimal.

## How the agent's five actions become physical commands

**C113** **Action decoding (each action a<sub>i</sub> in −1..1)**

Formula: $\text{throttle} = (a_0+1)/2$; $\theta = \pi\, a_1$; $\varphi = (\pi/2)\, a_2$; $u = u_{min} + (a_3+1)/2\,(u_{max}-u_{min})$; $S_{set} = 1 + 0.5\, a_4$

Values: $u_{min}$ = 20,000 m/s, $u_{max} = 5.00\times10^6$ m/s

Result: **packet mass = throttle × $m_{pk,max}$; direction ($\sin\theta\cos\varphi$, $\cos\theta\cos\varphi$, $\sin\varphi$) in radial/along-track/normal axes; mirror set-point 0.5-1.5 × today's sunlight**

Meaning: The agent never sets Earth's velocity; it only commands packets and mirror aim, and physics does the rest.

code: OctoEarthEnv.decode_action()

**C114** **Reward per decision (scoring only)**

Formula: $r = w_p\Delta\Phi - w_e(\text{e excess}) - w_i(\text{i excess}) - w_E E_{used}/E_0 - w_m m_{used}/m_0 - w_t - w_{Mars}\cdot\text{pen} - w_{clim}\cdot\text{pen}$ + terminal terms

Values: $w_p$=100, $w_E$=10, $w_m$=5, $w_t$=0.02, $w_{Mars}$=20, $w_{clim}$=2, success 100, failure 100

Result: **a number for learning only; all physical results are reported separately**

code: OctoEarthEnv.step()

# Calculation status, corrections, and limits

This page preserves the important cautions while removing the duplicate instrument summary. It distinguishes arithmetic checks from claims that need additional data or validation.

| Item | Required interpretation / missing information |
|---|---|
| **C5, C7, C16** | C5 is an ideal slow, quasi-circular Δv estimate; C7 holds total moving mass fixed; C16 is a rate-based minimum-time screen. None is an exact finite-mission trajectory result. |
| **C18 launch power** | The printed value is average power for the assumed yearly launch rate, not instantaneous pulse peak. Launch duration, storage, discharge power and acceleration profile are not given. |
| **C39 total impulse → Δv** | A scalar sum of local tangential impulses divided by initial mass is not a general orbital Δv calculation: direction rotates and system mass falls. Verify Δa/e/i and energy from the recorded trajectory instead. |
| **C81 beam momentum** | The report estimates E/c (≈114 m/s if perfectly aligned) but does not apply it. Actual effect needs direction-resolved photon momentum and transmitter recoil; the trajectory is not fully momentum-accounted without it. |
| **C110 energy closure** | A ledger residual near roundoff means the listed terms balance; it does not validate omitted physics. The reported $1.53\times10^{-6}$ mechanical-energy residual needs timestep-refinement evidence. |
| **Mission/PPO “success”** | The supplied PDF does not include matching raw logs, checkpoint or source for its claimed run. The results table also omits final inclination, Moon-bound status and post-mission stability outcome. Treat the result table as unverified, not independently reproduced. |
| **Climate outputs** | The two-box climate/albedo/latitude model is a toy energy-balance indicator. It cannot establish habitability, climate safety, “livable years,” or life-supporting conditions. |

## Important unresolved physical assumptions retained from source pp. 8–26

| | |
|---|---|
| **Arm heat and Earth damage** | C45–C47 assume a $10^{-9}$ heat leak without a radiator geometry. P6 assumes launch damage is harmless despite $\sim3\times10^{30}$ J per strategy-speed packet. Neither is demonstrated. |
| **Planets and long-term stability** | The source acknowledges omitted planets (including Jupiter) and does not establish long-term stability of the final 2.7 AU orbit. |
| **Other omitted climate/orbit effects** | Spin, seasons, tides, reflected-light distribution and real atmospheric/ocean response are not resolved. |

Read “FEASIBLE” only as “the simplified analytic screen passed under the stated hypothetical assumptions.” The revised C121 note gives the circular-aperture correction; the C122 note explains RK4 and the need for timestep convergence. Neither note turns these screening calculations into an engineering feasibility result.

# 10. What happens without each instrument

Each instrument is necessary for a reason. The calculations below show what would happen without it, or if it performed worse, using the same model functions. This is why each one is part of an enabler.

**C115** **Without the asteroid reservoir**

Formula $\Delta v = v_\infty \ln(m_{sys} / (m_{sys} - m_{thrown}))$, $m_{thrown} = 0$

Result **Δv = 0: Earth cannot change its orbit (momentum is conserved; there is nothing to throw)**

code: feasibility()

**C116** **Launcher limited to 30 km/s (fastest electromagnetic launchers studied)**

Formula $\Delta v_{cap} = \sqrt{(u^2 - v_{esc}^2)} \ln MR$

Values u = 30 km/s, MR = 1.004962

Result **138 m/s = 1.18 % of the required 11,658 m/s**

code: feasibility(max_ejection_speed_ms=3e4)

**C117** **Without the energy reserve (energy_budget_j = 0)**

Formula $E_{avail} = P_{elec,min}\ T$ vs $E_{min}$

Values $1.11\times10^{25} \times 3.16\times10^{11}$ vs $1.66\times10^{35}$

Result **still enough: $3.52\times10^{36}$ J available. At year 0 (30 % of the belt built) the arms get $3.87\times10^{24}$ W vs $1.56\times10^{24}$ W at full throttle with the strategy speed**

Meaning In the tested mission the reserve was not touched (all launch energy came from sunlight through E2). The reserve is an assumed store of energy with no source in the model; the mission does not need it.

code: feasibility(energy_budget_j=0)

**C118** **Radiators that are not directional (~8 % of the heat reaches Earth)**

Formula $T = [(S(1-\alpha)/4 + q/(4\pi R^2)) / (\varepsilon\sigma)]^{1/4}$

Values $q = 0.08 \times 1.05\times10^{24}$ W

Result **T ≈ 8,296 K: Earth's surface would be destroyed**

code: climate balance (indicative)

**C119** **Without the collector belt**

Formula $E_{avail}$ = reserve only

Values $3.00\times10^{35}$ J vs $1.66\times10^{35}$ J

Result **passes only because of the assumed reserve; Earth's own sunlight for the whole mission is only $5.48\times10^{28}$ J = $3.29\times10^{-7}$ of the minimum**

Meaning Without E2 the launch energy has no physical source; the belt is what makes the energy come from sunlight.

code: feasibility(solar_collectors=False)

**C120** **Transmitters 10 times smaller (3 km)**

Formula $\eta_{cap} = 1 - e^{-\tau^2}$, $\tau = D_t D_r/(\lambda d)$

Values $D_t$ = 3 km (only the product $D_t D_r$ matters, so this equals a 10 times smaller rectenna)

Result **capture 16.8 % at 1 AU, 2.1 % at 2.7 AU (instead of 99.9 % / 86.3 %); the rest is spilled**

code: collector_power()

### C121 Rectennas 10 times smaller (100 km)

Formula: same Goubau relation

Values: $D_r$ = 100 km

Result: **capture 16.8 % at 1 AU, 2.1 % at 2.7 AU: arm electricity $2.72\times10^{23}$ W at the target**

code: collector_power()

### C122 Without station keeping (tested, same mission)

Formula: RK4 orbits of 12 collectors without the sail-tilt control

Values: 3,600 simulated years

Result: **largest drift $1.46\times10^{8}$ km = 2,244 collector spacings (with it: 723 km): the belt would scramble, beams and lanes could not be kept**

code: BeltKeeper (check 19)

### C123 Without the clear lane

Formula: shade = 1 when |z|/r < f (Earth's line of sight to the Sun crosses the band)

Values: Earth in the ecliptic: |z|/r ≈ 0 < 0.3

Result: **shade = 1: Earth and its mirrors get no direct sunlight; only internal heat (~0.087 W/m²) would keep the surface near 40 K**

code: earth_shading()

### C124 Without factories (belt stays at 30 %)

Formula: $P_{elec}(b = b_0)$

Values: $b_0$ = 0.3

Result: **$3.87\times10^{24}$ W at 1 AU, $3.34\times10^{24}$ W at 2.7 AU (still above the tested mean demand $1.82\times10^{24}$ W, but aged panels would not be replaced: after 50 years the panels reach their end of life)**

code: collector_power()

### C125 Without mirrors at the target

Formula: equilibrium at $S = S_{1AU}/a_t^2$

Values: 0.1372 × today

Result: **152.4 K: snowball Earth (ice to the equator, no growing area)**

code: climate_equilibrium()

### C126 Pointing error larger than Earth's angular size from the ring

Formula: $\delta\theta > R_{Earth}/r$

Values: 0.0500 rad

Result: **the mirror light misses Earth: same as having no mirrors (C125)**

code: indicative

# C121 — Goubau relation and beam-capture calculation

The Goubau relation estimates the geometric fraction of radiated power intercepted by a receiving aperture. It is not the full electrical-to-electrical efficiency.

## 1. Link parameter

The area-based expression is $\tau^2 = A_t A_r/(\lambda d)^2$ and $\eta_{beam} \approx 1 - \exp(-\tau^2)$, where $A_t$ and $A_r$ are aperture areas, $\lambda = c/f$ is wavelength, and $d$ is instantaneous separation. For circular apertures with physical diameters $D_t, D_r$, $A = \pi D^2/4$, so:

$$\tau = (\pi/4)\, D_t D_r / (\lambda d)$$

The PDF instead uses $\tau = D_t D_r/(\lambda d)$, while defining circular apertures elsewhere. Unless it intended a different effective-aperture convention, the $\pi/4$ factor is missing; beam capture and downstream power should be recalculated.

## 2. Effect on C121 under the report geometry

For 5.8 GHz, $\lambda \approx 0.05169$ m. With collectors at 0.5 AU and Earth at 2.7 AU, the distance varies across the belt. A uniform latitude/longitude average over the stated 30%-of-sphere band gives the conditional comparison below.

| 2.7 AU link | PDF result | Circular-area estimate |
|---|---|---|
| **30 km Tx / 1,000 km Rx (C71)** | 86.3% capture | ≈71.6% |
| **30 km Tx / 100 km Rx (C121)** | 2.1% capture | ≈1.31% |
| **Arm electricity at full belt (C121)** | $2.72\times10^{23}$ W | $\approx 1.69\times10^{23}$ W |

Power relation: $P_{arms} = P_{PV}\eta_{mw}\bar{\eta}_{beam}\eta_r = (1.90\times10^{25}\ \text{W})(0.80)(\bar{\eta}_{beam})(0.85)$. C120's 3 km/1,000 km link has the same aperture-diameter product as C121's 30 km/100 km link, so the same geometric capture estimate applies. These averages are conditional independent calculations, not a reproduction of the PDF's missing 6,480-point implementation.

The exponential curve is approximate; accurate capture needs the field pattern over the receiver and realistic aperture illumination, sidelobes, pointing, polarization, blocking and hardware losses. Source: N. Shinohara, "Beam Efficiency of Wireless Power Transmission via Radio Waves from Short Range to Long Range" ($\tau^2 = AtAr/(\lambda d)^2$; the $\eta \approx 1 - e^{-\tau^2}$ relation is described as an approximation).

# C122 — What “RK4 orbits” means

RK4 describes how the equations of motion are numerically integrated. It is not a physical force or station-keeping technology.

## 1. Orbit equations and force model

A collector state typically obeys dr/dt=v and dv/dt=a(r,v,t). The model must specify a: gravity, radiation pressure, beam recoil, neighbor forces, and any controller acceleration. “With station keeping” and “without” should use the same initial conditions and physical forces, differing only by the controller input. RK4 advances whichever equations are supplied; it cannot fix missing or inaccurate forces.

## 2. Classical fourth-order Runge–Kutta step

For dy/dt=f(t,y) and step h, RK4 computes four slopes and combines them:

$$k_1=f(t_n,y_n);\ k_2=f(t_n+h/2,y_n+h\,k_1/2);$$
$$k_3=f(t_n+h/2,y_n+h\,k_2/2);\ k_4=f(t_n+h,y_n+h\,k_3);$$
$$y_{n+1}=y_n+(h/6)(k_1+2k_2+2k_3+k_4)$$

For smooth equations, RK4 has global truncation error of order $h^4$ (local error $h^5$). It is not symplectic, so long orbital runs can accumulate energy/angular-momentum drift; timestep refinement is necessary.

## 3. Interpreting the reported 3,600-year comparison

At a fixed half-day step, 3,600 years is about 2.63 million integration steps per run. The reported drift (723 km with station keeping; $1.46\times10^8$ km without) depends on initial conditions, force and controller laws, and numerical resolution—not on “RK4” alone. The PDF does not include enough trajectory data to independently reproduce those values.

| Needed to validate C122 | Reason |
|---|---|
| **Initial states and 12-point layout** | Sets collector phases and distances. |
| **All forces and controller/saturation equations** | Defines the actual orbit model. |
| **Half-day vs quarter-day (or finer) runs** | Shows the drift result converges. |
| **Energy/angular momentum and slot-error histories** | Exposes numerical drift and excursions. |

Do not extrapolate 12 sample collectors to five million without a representativeness argument and fleet-level checks.

# 13. Appendix: every parameter

| Key | Meaning | Value | Unit | Category |
|---|---|---|---|---|
| asteroid_mass_kg | Initial asteroid mass | 3e+22 | kg | Engineering assumption |
| max_packet_mass_kg | Max packet mass | 1e+18 | kg | Engineering assumption |
| max_ejection_speed_ms | Max ejection speed (exhaust velocity) | 5000000.0 | m/s | Engineering assumption |
| min_ejection_speed_ms | Min ejection speed | 20000.0 | m/s | Engineering assumption |
| energy_budget_j | Propulsion energy budget | 3e+35 | J | Engineering assumption |
| launch_efficiency | Launch efficiency | 0.5 | 0-1 | Engineering assumption |
| ejections_per_year | Ejection frequency | 8.0 | 1/yr | Engineering assumption |
| target_au | Target orbit radius | 2.7 | AU | Mission requirement |
| mission_years | Mission duration | 10000.0 | yr | Mission requirement |
| decision_interval_years | Decision interval | 100.0 | yr | Simulation setting |
| tol_a_au | Tolerance on target radius | 0.05 | AU | Mission requirement |
| tol_e | Tolerance on eccentricity | 0.05 | - | Mission requirement |
| tol_inc_deg | Tolerance on inclination | 5.0 | deg | Mission requirement |
| validate_years | Stability check duration | 200.0 | yr | Mission requirement |
| randomize_phase | Randomize starting position | True | on/off | Simulation setting |
| train_episodes | Training episodes | 1000 | - | RL hyperparameter |
| ppo_n_steps | PPO steps per update | 1000 | - | RL hyperparameter |
| ppo_batch_size | PPO mini-batch size | 200 | - | RL hyperparameter |
| ppo_epochs | PPO epochs per update | 10 | - | RL hyperparameter |
| ppo_learning_rate | Learning rate | 0.0003 | - | RL hyperparameter |
| ppo_gamma | Discount factor (gamma) | 0.995 | - | RL hyperparameter |
| eval_every_episodes | Evaluate every | 50 | episodes | RL hyperparameter |
| seed | Random seed | 1 | - | Simulation setting |
| w_progress | Reward: progress weight | 100.0 | points | Reward weight (scoring only) |
| w_eccentricity | Reward: eccentricity penalty | 1.0 | points | Reward weight (scoring only) |
| w_inclination | Reward: inclination penalty | 1.0 | points | Reward weight (scoring only) |

| | | | | |
|---|---|---|---|---|
| w_energy | Reward: energy-use penalty | 10.0 | points | Reward weight (scoring only) |
| w_mass | Reward: mass-use penalty | 5.0 | points | Reward weight (scoring only) |
| w_time | Reward: time penalty | 0.02 | points/decision | Reward weight (scoring only) |
| w_success | Reward: success bonus | 100.0 | points | Reward weight (scoring only) |
| w_terminal_error | Reward: final-miss penalty | 50.0 | points | Reward weight (scoring only) |
| w_failure | Reward: failure penalty | 100.0 | points | Reward weight (scoring only) |
| w_mars_safety | Reward: Mars-proximity penalty | 20.0 | points | Reward weight (scoring only) |
| mars_gravity | Mars gravity on | True | true/false | Planetary safety (Mars) |
| mars_max_tide_ratio | Max Mars tide on Earth (x Moon's tide) | 0.01 | - | Planetary safety (Mars) |
| mars_max_moon_perturbation | Max Mars pull on the Moon's orbit | 0.001 | - | Planetary safety (Mars) |
| mars_min_distance_km | Hard minimum Earth-Mars distance | 2000000.0 | km | Planetary safety (Mars) |
| mars_shield | Mars safety shield | True | true/false | Planetary safety (Mars) |
| shield_caution_factor | Shield caution factor | 2.0 | x safe distance | Planetary safety (Mars) |
| shield_horizon_years | Shield forecast horizon | 10.0 | yr | Planetary safety (Mars) |
| solar_collectors | Solar collectors on | True | true/false | Solar collectors (energy supply) |
| collector_fraction | Share of the Sun's output collected | 0.3 | 0-1 | Solar collectors (energy supply) |
| collector_orbit_au | Collector belt distance from the Sun | 0.5 | AU | Solar collectors (energy supply) |
| collector_count | Number of collectors | 5000000.0 | - | Solar collectors (energy supply) |
| collector_areal_density | Collector mass per area | 0.1 | kg/m^2 | Solar collectors (energy supply) |
| pv_efficiency | Solar-cell efficiency | 0.25 | 0-1 | Solar collectors (energy supply) |
| microwave_efficiency | Electricity-to-microwave efficiency | 0.8 | 0-1 | Solar collectors (energy supply) |
| microwave_freq_ghz | Microwave frequency | 5.8 | GHz | Solar collectors (energy supply) |
| transmitter_diameter_km | Transmitter antenna diameter | 30.0 | km | Solar collectors (energy supply) |
| rectenna_diameter_km | Rectenna diameter (per arm) | 1000.0 | km | Solar collectors (energy supply) |
| rectenna_efficiency | Rectenna efficiency | 0.85 | 0-1 | Solar collectors (energy supply) |

| | | | | |
|---|---|---|---|---|
| pv_temp_coeff | Solar-cell temperature coefficient | 0.002 | 1/K | Solar collectors (energy supply) |
| collector_absorptance | Panel absorptance | 0.9 | 0-1 | Solar collectors (energy supply) |
| collector_emissivity | Panel emissivity | 0.85 | 0-1 | Solar collectors (energy supply) |
| pv_degradation_per_year | Panel degradation | 0.002 | 1/yr | Solar collectors (energy supply) |
| collector_lifetime_years | Panel lifetime | 50.0 | yr | Solar collectors (energy supply) |
| collector_recycle_frac | Recycled share of replaced panels | 0.95 | 0-1 | Solar collectors (energy supply) |
| collector_initial_built | Belt built at year 0 | 0.3 | 0-1 | Solar collectors (energy supply) |
| collector_build_years | Build-up time of the belt | 400.0 | yr | Solar collectors (energy supply) |
| collector_embodied_energy_jkg | Embodied energy of collectors | 100000000.0 | J/kg | Solar collectors (energy supply) |
| earth_lane_clear | Keep Earth's sunlight lane clear | True | true/false | Solar collectors (energy supply) |
| collector_sk_authority | Station-keeping authority | 0.5 | 0-1 | Solar collectors (energy supply) |
| climate_model | Climate model on | True | true/false | Climate + space mirrors |
| mirrors | Space mirrors on | True | true/false | Climate + space mirrors |
| climate_target_K | Target surface temperature | 288.0 | K | Climate + space mirrors |
| climate_tol_K | Temperature tolerance | 5.0 | K | Climate + space mirrors |
| mirror_area_m2 | Total mirror area | 16000000000000000.0 | m^2 | Climate + space mirrors |
| mirror_reflectivity | Mirror reflectivity | 0.9 | 0-1 | Climate + space mirrors |
| mirror_orbit_re | Mirror orbit radius | 20.0 | Earth radii | Climate + space mirrors |
| mirror_areal_density | Mirror film mass per area | 0.005 | kg/m^2 | Climate + space mirrors |
| arm_heat_to_earth_frac | Arm waste heat reaching Earth | 1e-09 | 0-1 | Climate + space mirrors |
| beam_spill_to_earth_frac | Beam spill reaching Earth | 1e-09 | 0-1 | Climate + space mirrors |
| w_climate | Reward: climate penalty | 2.0 | points per 100 y | Reward weight (scoring only) |
| hifi_dt_hours | High-fidelity time step | 6.0 | h | Simulation setting |
| max_fragments | Max tracked fragments | 48 | - | Simulation setting |
| release_radius_m | Packet hand-over distance | 2000000000.0 | m | Simulation setting |

| | | | | |
|---|---|---|---|---|
| `fragment_escape_au` | Fragment escape distance | 60.0 | AU | Simulation setting |

130 calculations. End of report.